

1 Osnove teorije

1.1 BCS teorija pri končni temperaturi

BCS model, kjer je $\xi_{\mathbf{k},\sigma} = \epsilon_{\mathbf{k},\sigma} - E_F$ in $k := (\mathbf{k}, \uparrow)$, $-k := (-\mathbf{k}, \downarrow)$:

$$H = \sum_k \xi_k (c_k^\dagger c_k + c_{-k}^\dagger c_{-k}) - V \sum_{k,k'} c_k^\dagger c_{-k}^\dagger c_{-k'} c_{k'} \quad (1)$$

Naredimo približek povprečnega polja, definiramo $b_k = c_k c_{-k}$ in $b_k = \langle b_k \rangle + \delta b_k = F_k + \delta b_k$, sledi (do $\mathcal{O}(\delta b_k^2)$)

$$b_k^\dagger b_{k'} \simeq F_k^* F_{k'} + F_k^* \delta b_{k'} + F_{k'} \delta b_k^\dagger = F_k^* b_{k'} + F_{k'} b_k^\dagger - F_k^* F_{k'} = \langle b_k^\dagger \rangle b_{k'} + b_k^\dagger \langle b_{k'} \rangle - \langle b_k^\dagger \rangle \langle b_{k'} \rangle \quad (2)$$

Iz česar sledi:

$$H = \sum_k (c_k^\dagger \quad c_{-k}) \begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta_k^* & -\xi_k \end{pmatrix} \begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} + \frac{|\Delta|^2}{V} + \sum_k \xi_k \quad (3)$$

Ob definiciji $\Delta_k = \sum_{k'} V(k - k') \langle c_{k'} c_{-k'} \rangle$, vzamemo še lokalno interakcijo, zato je $V(q) \simeq V$ in $\Delta_k \simeq \Delta$. Naredimo Bogoljubov rotacijo, privzamemo $u_k, v_k \in \mathbb{R}$ in simetrijo okoli E_F : $u_k = u_{-k}$ ter $v_k = v_{-k}$. Iz CCR sledi $u_k^2 + v_k^2 = 1$. Kvazidelci so definirani z rotacijo

$$\begin{pmatrix} c_k \\ c_{-k}^\dagger \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix} \quad (4)$$

nov Hamiltonjan pa postane

$$H = \sum_k (\alpha_k^\dagger \quad \alpha_{-k}) \begin{pmatrix} \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) - \xi_k v_k^2 & -2\xi_k u_k v_k + \Delta_k u_k^2 - \Delta_k^* v_k^2 \\ -2\xi_k u_k v_k - \Delta_k v_k^2 + \Delta_k^* u_k^2 & \xi_k v_k^2 - \xi_k u_k^2 + u_k v_k (\Delta_k + \Delta_k^*) \end{pmatrix} \begin{pmatrix} \alpha_k \\ \alpha_{-k}^\dagger \end{pmatrix}, \quad (5)$$

kjer zahtevamo, da so izvendiagonalni členi enaki 0. Iz CCR zahteve lahko vzamemo parameterizacijo (tu sem vzel kompleksno nastavimo, da je $\Delta = \Delta^* e^{i2\phi_k}$):

$$v_k = \sin\left(\frac{\theta_k}{2}\right) e^{i\phi_k} \quad (6)$$

$$u_k = \cos\left(\frac{\theta_k}{2}\right) \quad (7)$$

$$u_k v_k = \cos\left(\frac{\theta_k}{2}\right) \sin\left(\frac{\theta_k}{2}\right) = \frac{1}{2} \sin \theta_k \quad (8)$$

$$u_k^2 - v_k^2 = \cos^2\left(\frac{\theta_k}{2}\right) - \sin^2\left(\frac{\theta_k}{2}\right) = \cos \theta_k \quad (9)$$

skupaj dobimo (od tu naprej $\Delta := |\Delta|$):

$$\xi_k \sin(\theta_k) = \Delta \cos(\theta_k) \quad (10)$$

kar fiksira

$$\tan(\theta_k) = \frac{|\Delta_k|}{\xi_k}. \quad (11)$$

Tako iz trikotnika preberemo

$$\sin \theta_k = \frac{|\Delta_k|}{E_k} = \frac{|\Delta_k|}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (12)$$

$$\cos \theta_k = \frac{\xi_k}{E_k} = \frac{\xi_k}{\sqrt{\xi_k^2 + |\Delta_k|^2}} \quad (13)$$

$$(14)$$

kjer je $E_k = \pm \sqrt{\xi_k^2 + |\Delta_k|^2}$ energija kvazidelcev:

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k - E_k \alpha_{-k}^\dagger \alpha_{-k} + \xi_k + \frac{|\Delta|^2}{V}. \quad (15)$$

V termičnem stanju so diagonalni delci porazdeljeni po

$$\langle \alpha_k^\dagger \alpha_k \rangle = \frac{1}{1 + \exp(E_k/T)} = \frac{1}{2} - \frac{1}{2} \tanh\left(\frac{E_k}{2T}\right). \quad (16)$$

Uporabni sta zvezi $f(-E/T) = 1 - f(E/T)$ in $1 - 2f(E/T) = \tanh(E/(2T))$. V definiciji ureditvenega parametra je

$$\langle c_k c_{-k} \rangle = \left\langle \left(u_k \alpha_k - v_k \alpha_{-k}^\dagger \right) \left(u_k \alpha_{-k} + v_k \alpha_k^\dagger \right) \right\rangle = u_k v_k \left(1 - \langle \alpha_{-k}^\dagger \alpha_{-k} \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = \frac{1}{2} \sin \theta_k \tanh\left(\frac{\beta E_k}{2}\right) \quad (17)$$

iz česar sledi enačba za režo pri $T > 0$ (seštevamo po k okoli E_F):

$$\Delta = \frac{V}{2} \sum_k' \sin \theta_k \tanh\left(\frac{E_k}{2T}\right) = \frac{V\Delta}{2} \sum_k \frac{\tanh\left(\frac{1}{2T} \sqrt{\Delta^2 + \xi_k^2}\right)}{\sqrt{\Delta^2 + \xi_k^2}} \quad (18)$$

1.2 Minimalni model ekscitonskega izolatorja pri končni temperaturi

Pri izolatorju za minimalni model vzamemo dve orbitali (a, b) model tesne vezi v vsaki posebej in lokalno Coulombovo interakcijo na mestu j med elektronskima gostotama v posameznih pasovih

$$H = H_0 + H_V = \sum_{i,j} \left(\epsilon_a^{(0)} \delta_{ij} + t_{ij}^{(a)} \right) c_{a,i}^\dagger c_{a,j} + \sum_{i,j} \left(\epsilon_b^{(0)} \delta_{ij} + t_{ij}^{(b)} \right) c_{b,i}^\dagger c_{b,j} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (19)$$

$$= \sum_k \epsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \epsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (20)$$

najprej diagonaliziramo vsako orbitalo posebej, kar da gole disperzije (vezavni energiji nastavimo na $\pm G/2$) (delamo v dveh dimenzijah in mrežna konstanta $a_0 = 1$):

$$\epsilon_{a(b),k} = \pm \frac{G}{2} - 2t_{a(b)} (\cos(k_x) + \cos(k_y)), \quad (21)$$

v tem primeru je najmanjša razdalja med pasovoma $G_0 = G - 4(t_a - t_b)$ oziroma $G_0 = G - 8t_a$, če je $t_b = -t_a$. Za interakcijo spet naredimo približek povprečnega polja:

$$c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \approx \langle c_{a,j}^\dagger c_{a,j} \rangle c_{b,j}^\dagger c_{b,j} + \langle c_{b,j}^\dagger c_{b,j} \rangle c_{a,j}^\dagger c_{a,j} - \langle c_{a,j}^\dagger c_{b,j} \rangle c_{b,j}^\dagger c_{a,j} - \langle c_{b,j}^\dagger c_{a,j} \rangle c_{a,j}^\dagger c_{b,j} + \text{konst.} \quad (22)$$

od tod sledi

$$H_V^{\text{MF}} = V n_b \sum_j c_{a,j}^\dagger c_{a,j} + V n_a \sum_j c_{b,j}^\dagger c_{b,j} - V \phi \sum_j c_{b,j}^\dagger c_{a,j} - V \phi^* \sum_j c_{a,j}^\dagger c_{b,j} + NV |\phi|^2 \quad (23)$$

kjer smo definirali:

$$n_a = \langle c_{a,j}^\dagger c_{a,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{a,k} \rangle \quad (24)$$

$$n_b = \langle c_{b,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{b,k}^\dagger c_{b,k} \rangle \quad (25)$$

$$\phi = \langle c_{a,j}^\dagger c_{b,j} \rangle = \frac{1}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle \quad (26)$$

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle, \quad \Delta^* = -\frac{V}{N} \sum_k \langle c_{b,k}^\dagger c_{a,k} \rangle \quad (27)$$

$$\Delta_k = \frac{1}{N} \sum_{k'} V(k - k') \langle a_k^\dagger, b_{k'} \rangle, \quad V(q) = \int e^{-i\mathbf{q} \cdot \mathbf{r}} V(\mathbf{r}) \, d\mathbf{r} \quad (28)$$

Uporabimo FT še na celotnem Hamiltonjanu

$$c_{a,j}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{-i\mathbf{k} \cdot \mathbf{r}_j} c_{a,k}^\dagger \quad (29)$$

$$H^{\text{MF}} = \sum_k \left([\epsilon_{a,k} + V n_b] c_{a,k}^\dagger c_{a,k} + [\epsilon_{b,k} + V n_a] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) + \left(\frac{|\Delta|^2}{V} N - V N n_a n_b \right) \quad (30)$$

oziroma:

$$H^{\text{MF}} = \sum_k \begin{pmatrix} c_{a,k}^\dagger & c_{b,k}^\dagger \end{pmatrix} \begin{pmatrix} \epsilon_{a,k}^H & \Delta^* \\ \Delta & \epsilon_{b,k}^H \end{pmatrix} \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} + \text{konst.}, \quad (31)$$

kjer definiramo Hartree energiji $\epsilon_{a(b),k}^H = \epsilon_{a(b),k} + V n_{b(a)}$. Transformacija v diagonalno / Bogoljubov bazo je podobna kot pri BCS, le da ne meša kreatorjev in anihilatorjev / ohranja število delcev:

$$\begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix} \quad (32)$$

od tod po enakem postopku kot prej dobimo energiji kvazidelcev

$$E_{\alpha(\beta)}(k) = \frac{\epsilon_{a,k}^H + \epsilon_{b,k}^H}{2} \pm \sqrt{\left(\frac{\epsilon_{a,k}^H - \epsilon_{b,k}^H}{2} \right)^2 + |\Delta|^2} := \eta_k \pm \sqrt{\xi_k^2 + \Delta^2} \quad (33)$$

Za bookkeeping:

$$\left(\frac{\epsilon_a^H - \epsilon_b^H}{2} \right) \sin \theta_k + \Delta \cos \theta_k = 0 \implies \tan \theta_k = \frac{-2\Delta}{\epsilon_a^H - \epsilon_b^H} = \frac{-\Delta}{\xi_k} \quad (34)$$

$$u_k v_k = -\frac{\sin \theta_k}{2}. \quad (35)$$

Za samousklajeno enačbo zdaj potrebujemo

$$\langle c_{a,k}^\dagger c_{b,k} \rangle = \langle (u_k \alpha_k^\dagger + v_k \beta_k^\dagger)(-v_k \alpha_k + u_k \beta_k) \rangle = u_k v_k \left(\langle \beta_k^\dagger \beta_k \rangle - \langle \alpha_k^\dagger \alpha_k \rangle \right) = -\frac{1}{2} \sin \theta_k (f(E_\beta) - f(E_\alpha)) \quad (36)$$

od tod

$$\Delta = -\frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle = \frac{V}{N} \sum_k \frac{\Delta}{2\sqrt{\xi_k^2 + |\Delta|^2}} (f(E_\beta) - f(E_\alpha)) \quad (37)$$

Opomba, lahko definiramo tudi $n = n_a + n_b = \text{konst.} = 1$ in $n_a - n_b =: m$ od tod sledi

$$H^{\text{MF}} = \sum_k \left(\left[\epsilon_{a,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{a,k}^\dagger c_{a,k} + \left[\epsilon_{b,k} + \frac{V}{2} n - \frac{V}{2} m \right] c_{b,k}^\dagger c_{b,k} + \Delta^* c_{a,k}^\dagger c_{b,k} + \Delta c_{b,k}^\dagger c_{a,k} \right) \quad (38)$$

od koder je v 2D Hartree gap enak:

$$G^H = G - 4(t_a - t_b) - Vm. \quad (39)$$

Note: v praksi je $m < 0$ V dveh dimenzijah se Hartree pasova srečata na

$$\mu_B = \frac{V}{2} n - \frac{t_a + t_b}{t_a - t_b} \frac{G - Vm}{2}. \quad (40)$$

Zamik je enak $V/2$, ko sta osnovna pasova simetrična in je zasedenost polovična. V primeru simetričnih pasov sta disperziji kvazidelcev

$$E_{\alpha(\beta)}(k) - V/2 = \pm \sqrt{\left(\frac{\epsilon_{a,k} - \epsilon_{b,k} - Vm}{2}\right)^2 + |\Delta|^2}. \quad (41)$$

Velja še

$$n_a = \frac{1}{N} \sum_k (u_k^2 f(E_\alpha) + v_k^2 f(E_\beta)) \quad (42)$$

$$n_b = \frac{1}{N} \sum_k (v_k^2 f(E_\alpha) + u_k^2 f(E_\beta)) \quad (43)$$

$$u_k^2 = \frac{1}{2} \left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (44)$$

$$v_k^2 = \frac{1}{2} \left(1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}} \right) \quad (45)$$

1.3 Numerika/psevdokoda

1.4 Odprti sistemi in modeliranje kopeli

Born-Markov master enačbe in Redfield: ločimo kopel in sistem,

$$\rho_S(t) = \text{Tr}_B(\rho(t)) \quad (46)$$

Hamiltonjan razcepimo na $H = H_0 + H_I$, kjer je $H_0 = H_S \otimes \mathbb{I}_B + \mathbb{I}_S \otimes H_B$ razdeljen na podсистema, za interakcijski del pa privzamemo obliko $H_I = \sum_\alpha A_\alpha \otimes B_\alpha$. V interakcijski sliki je operator za evolucijo

$$U_0(t, t_0) = e^{-iH_0(t-t_0)} \quad (47)$$

$$\rho_I(t) = U_0^\dagger(t, 0) \rho(t) U_0(t, 0) = e^{iH_0 t} \rho(t) e^{-iH_0 t} \quad (48)$$

$$H_I(t) = U_0^\dagger(t, 0) H_I U_0(t, 0) = e^{iH_0 t} H_I e^{-iH_0 t} \quad (49)$$

Von Neumannova enačba je

$$\frac{d}{dt} \rho_I(t) = -i[H_I(t), \rho_I(t)] \quad (50)$$

$$\rho_I(t) = \rho_I(0) - i \int_0^t dt' [H_I(t'), \rho_I(t')] \quad (51)$$

vstavimo nazaj v komutator in se znebimo prostosnih stopenj kopeli:

$$\frac{d}{dt} \rho_S(t) = - \int_0^t ds \text{Tr}_B ([H_I(t), [H_I(t-s), \rho_S(t) \otimes \rho_B]]) \quad (52)$$

V interakcijskem členu so operatorji A_α napisani kot lastni operatorji H_S z energijami prehodov $\omega = E_b - E_a$

$$A_\alpha(\omega) = \sum_{E_b - E_a = \omega} |a\rangle \langle a| A_\alpha |b\rangle \langle b| \quad (53)$$

To zadošča relacijam $[H_S, A_\alpha(\omega)] = -\omega A_\alpha(\omega)$, $[H_S, A_\alpha^\dagger(\omega)] = \omega A_\alpha^\dagger(\omega)$ in $e^{iH_S t} A_\alpha(\omega) e^{-iH_S t} = e^{-i\omega t} A_\alpha(\omega)$, $e^{iH_S t} A_\alpha^\dagger(\omega) e^{-iH_S t} = e^{i\omega t} A_\alpha^\dagger(\omega)$. V interakcijski sliki potem velja

$$H_I(t) = \sum_{\alpha, \omega} e^{-i\omega t} A_\alpha(\omega) \otimes B_\alpha(t) = e^{iH_B t} B_\alpha e^{-iH_B t} \quad (54)$$

evolucija sledi kot

$$\dot{\rho}_S = \sum_{\omega, \omega'} \sum_{\alpha, \beta} e^{i(\omega' - \omega)t} \Gamma_{\alpha\beta}(\omega) \left(A_\alpha(\omega) \rho_S(t) A_\beta^\dagger(\omega') - A_\beta^\dagger(\omega') A_\alpha(\omega) \rho_S(t) \right) + \text{h.c.} \quad (55)$$

kjer so prehodne matrike med stanji

$$\Gamma_{\alpha\beta}(\omega) = \int_0^\infty ds e^{i\omega s} \text{Tr}_B (B_\alpha^\dagger(t) B_\beta(t-s) \rho_B) = \frac{1}{2} \gamma_{\alpha\beta}(\omega) + i S_{\alpha\beta}(\omega) \quad (56)$$

Evolucija pa ima strukturo

$$\frac{d\rho_S}{dt} = -i[H_{LS}, \rho_S(t)] + \mathcal{D}(\rho_S(t)) \quad (57)$$

z disipatorjem

$$\mathcal{D}(\rho_S) = \sum_{\omega} \sum_{\alpha, \beta} \gamma_{\alpha\beta}(\omega) \left(A_\beta(\omega) \rho_S A_\alpha^\dagger(\omega) - \frac{1}{2} \{A_\alpha^\dagger(\omega) A_\beta(\omega), \rho_S\} \right) \quad (58)$$

oz. v splošnem

$$\mathcal{D}(\rho_S) = \sum_k \gamma_k \left(L_k \rho_S L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho_S\} \right) \quad (59)$$

V stacionarnem stanju velja

$$\mathcal{L}\rho_{\text{ness}} = 0. \quad (60)$$

Notes za vektorizacijo in superoperatorje:

$$\mathcal{L}u_\mu^R = \lambda_\mu u_\mu^R \quad (61)$$

$$\text{vec}(ABC) = (A \otimes C^T) \text{vec}(B) \quad (62)$$

$$\text{Tr}(A^\dagger B) = \text{vec}(A)^\dagger \text{vec}(B) \quad (63)$$

$$\frac{d}{dt}|\rho\rangle = \left(-iH \otimes \mathbb{I} + i\mathbb{I} \otimes H^T + \sum_k L_k \otimes L_k^* - \frac{1}{2} \sum_k (L_k^\dagger L_k \otimes \mathbb{I} + \mathbb{I} \otimes L_k^T L_k^*) \right) |\rho\rangle \quad (64)$$

$$\langle \hat{O} \rangle(t) = \langle \mathbb{I} | \hat{O} \otimes \mathbb{I} | \rho(t) \rangle \quad (65)$$

Šibke sklopitve in perturbacijski pristopi

$$\mathcal{L} = -i[H, \cdot] + \epsilon \mathcal{D}[\cdot], \quad \rho = \rho_0 + \epsilon \rho_1 + \epsilon^2 \rho_2 + \dots \quad (66)$$

napišemo gostotno matriko sistema v lastni bazi, siagonalno, s klasičnimi populacijami

$$\rho_0 = \sum_n p_n |n\rangle \langle n|, \quad H|n\rangle = E_n |n\rangle \quad (67)$$

kjer velja v ničtem redu (by construction)

$$\mathcal{L}\rho_0 = -i[H, \rho_0] = 0 \quad (68)$$

V prvem redu so enačbe za NESS

$$-i[H, \rho_1] + \mathcal{D}(\rho_0) = 0 \quad (69)$$

kjer je $\mathcal{D}(\rho_0) = \sum_{\mu} (L_{\mu}\rho_0 L_{\mu}^{\dagger} - \frac{1}{2}\{L_{\mu}^{\dagger}L_{\mu}, \rho_0\})$ in od tod enačba za prehode med populacijami

$$\langle n | \mathcal{D}(\rho_0) | n \rangle = \sum_{\mu} \sum_m p_m |\langle n | L_{\mu} | m \rangle|^2 - \sum_{\mu} \sum_m p_n |\langle m | L_{\mu} | n \rangle|^2 \quad (70)$$

kot klasična populacijska dinamika z $R_{nm} = \sum_{\mu} |\langle n | L_{\mu} | m \rangle|^2$

$$\dot{p}_n = \epsilon \sum_m (R_{nm} p_m - R_{mn} p_n) \quad (71)$$

Izven diagonali elementi se razvijejo hitro na $\mathcal{O}(1)$, medtem ko se populacije razvijajo počasi na GGE manifoldu z $\mathcal{O}(\epsilon^{-1})$. Za ohranitev normalizacije mora veljati $\sum_n \dot{p}_n = 0$.

Integrabilni sistemi imajo makroskopsko število ohranjenih nabojev $[H_0, C_i] = 0$, pod unitarno časovno evolucijo se razvijejo v netermično stanje

$$\rho_{GGE} = \frac{1}{Z} e^{-\sum_i \lambda_i C_i} \quad (72)$$

$$\langle \psi_0 | C_i | \psi_0 \rangle = \text{Tr}(C_i \rho_{GGE}) \quad (73)$$

fiksira λ_i . UNitarna dinamika pripcira na GGE manifold, šibka sklopitev premika po GGE manifoldu:

$$\rho_0(t) = \rho_{\lambda(t)} = \frac{1}{Z(t)} \exp \left(- \sum_i \lambda_i(t) C_i \right) \quad (74)$$

EOM za ohranjene naboje je

$$\langle \dot{C}_i \rangle_{\lambda(\infty)} = \text{Tr}[C_i \mathcal{D}\rho] \simeq \epsilon \text{Tr}[C_i \mathcal{D}\rho_0] = 0 \quad \forall \lambda_i \quad (75)$$

Definiramo projektor na GGE manifold:

$$PX = - \sum_{ij} \frac{\partial \rho}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j X] \quad (76)$$

ki da dinamiko vzdolž manifolda:

$$\frac{\partial \rho_{\lambda}}{\partial t} = \sum_i \frac{\partial \rho_{\lambda}}{\partial \lambda_i} \dot{\lambda}_i = \epsilon P \mathcal{D}(\rho_{\lambda}) = -\epsilon \sum_{ij} \frac{\partial \rho_{\lambda}}{\partial \lambda_i} (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}(\rho_{\lambda})] \quad (77)$$

od koder sledi evolucija za koeficiente/potenciale λ_i :

$$\dot{\lambda}_i = -\epsilon \sum_j (\chi^{-1})_{ij} \text{Tr}[C_j \mathcal{D}\rho_{\lambda(t)}] \quad (78)$$

s susceptibilnostmi $\chi_{ij} = \langle C_i C_j \rangle - \langle C_i \rangle \langle C_j \rangle = -\text{Tr} \left[C_i \frac{\partial \rho}{\partial \lambda_j} \right]$.

2 Ekscitonski izolatorji in sklopitev s kopelmi

$$\text{Tr} \left[f \frac{1}{2} \{\rho, A\} \right] = \text{Tr} \left[\frac{1}{2} f \rho A + \frac{1}{2} f A \rho \right] = \frac{1}{2} \text{Tr} [A f \rho + f A \rho] \quad (79)$$

$$= \frac{1}{2} \text{Tr} [(A f + f A) \rho] = \frac{1}{2} \langle A f + f A \rangle \quad (80)$$

2.1 Osnovni modeli kopeli

2.2 Limita neskončne temperature

2.3 Fononske sklopitve

Tu sta dve (tri) opcije. Najprej pogledamo, kako modificiran (Peierls tip) hopping integral in modificiran orbital energy vplivata na sklopitev. Model spremenimo: atoma znotraj o.c. sta na enakem položaju in se deformirata vedno v isto smer, torej $\mathbf{u}_{a,i} = \mathbf{u}_{b,i} = \mathbf{u}_i$:

$$H(\{\mathbf{u}\}) = \underbrace{\sum_{i,\gamma,\tau} \epsilon_{\gamma\tau}(\mathbf{u}_i) c_{\gamma,i}^\dagger c_{\tau,i}}_{\text{on-site in CEF}} + \underbrace{\sum_{\langle i,j \rangle, \gamma, \tau} t_{\gamma\tau}(\mathbf{r}_{ij}) c_{\gamma,i}^\dagger c_{\tau,j}}_{\text{hopping, med orbitalami}} + \underbrace{V \sum_j n_{a,j} n_{b,j}}_{\text{Coulomb}} \quad (81)$$

Predpostavimo, da se z deformacijo energija na orbitalo spremeni (podobnokot Kittel, str. 131), torej:

$$\begin{pmatrix} \epsilon_a^{(0)} & 0 \\ 0 & \epsilon_b^{(0)} \end{pmatrix} \rightarrow \begin{pmatrix} \epsilon_a^{(0)} + D_a \sum_{\langle ij \rangle} \hat{\mathbf{R}}_{ij}^0 \cdot (\mathbf{u}_j - \mathbf{u}_i) & D_{ab} \sum_{\langle ij \rangle} (\hat{R}_{ij}^0)_x \delta \mathbf{R}_{ij} \\ D_{ab} \sum_{\langle ij \rangle} (\hat{R}_{ij}^0)_x \delta \mathbf{R}_{ij} & \epsilon_b^{(0)} + D_b \sum_{\langle ij \rangle} \hat{\mathbf{R}}_{ij}^0 \cdot (\mathbf{u}_j - \mathbf{u}_i) \end{pmatrix} \quad (82)$$

$$\mathbf{r}_{\ell,j} = \mathbf{R}_j + \mathbf{u}_{\ell,j}. \quad (83)$$

$$\mathbf{U}_j = \frac{\mathbf{u}_{a,j} + \mathbf{u}_{b,j}}{2}, \quad \mathbf{r}_j = \mathbf{u}_{a,j} - \mathbf{u}_{b,j}. \quad (84)$$

$$\mathbf{u}_{a,j} = \mathbf{U}_j + \frac{\mathbf{r}_j}{2}, \quad \mathbf{u}_{b,j} = \mathbf{U}_j - \frac{\mathbf{r}_j}{2}. \quad (3)$$

Razvijemo

$$H_e[\mathbf{u}] = H_e[0] + H_{e-ph}^{(1)} + O(u^2), \quad H_{e-ph}^{(1)} = \sum_{\ell,j} \mathbf{u}_{\ell,j} \cdot \left. \frac{\partial H_e}{\partial \mathbf{u}_{\ell,j}} \right|_0. \quad (85)$$

$$t_{ij}^{(aa)}(\mathbf{r}_{ij}) \approx t_{ij}^{(a)0} + \alpha_t^{(a)} \hat{\mathbf{R}}_{ij}^0 \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (86)$$

$$t_{ij}^{(bb)}(\mathbf{r}_{ij}) \approx t_{ij}^{(b)0} + \alpha_t^{(b)} \hat{\mathbf{R}}_{ij}^0 \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (87)$$

$$t_{ij}^{(ab)}(\mathbf{r}_{ij}) \approx \underbrace{0}_{\text{ni hoppinga v nedeform. mreži}} + \alpha_t^{(ab)} \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (88)$$

$$\alpha_t^{(ab)} = \nabla_{\mathbf{r}} t_{ij}^{(ab)} \Big|_{\mathbf{R}_{ij}^0} \neq 0 \quad (89)$$

$$H_e = \sum_j \left(\epsilon_a c_{ja}^\dagger c_{ja} + \epsilon_b c_{jb}^\dagger c_{jb} \right) - \sum_{\langle i,j \rangle} \left(t_a(\mathbf{R}_j - \mathbf{R}_i) c_{ja}^\dagger c_{ia} + t_b(\mathbf{R}_j - \mathbf{R}_i) c_{jb}^\dagger c_{ib} + \text{h.c.} \right) + \sum_{\langle i,j \rangle} t_{ab}(\mathbf{R}_j - \mathbf{R}_i) c_{ja}^\dagger c_{ib} + \text{h.c.} \quad (90)$$

$$\mathbf{d}_{ij,ab} = \mathbf{r}_{j,b} - \mathbf{r}_{i,a} = (\mathbf{R}_j - \mathbf{R}_i) + (\mathbf{r}_{jb} - \mathbf{r}_{ia}) + (\mathbf{u}_{j,b} - \mathbf{u}_{i,a}) \quad (91)$$

$$t_a = t_a(d_{ij,aa}), \quad t_b = t_b(d_{ij,bb}) \quad (92)$$

$$t_a = t_a(d_{ij,aa}^0) + \nabla_{\mathbf{r}} t_a(\mathbf{r}) \Big|_{\mathbf{d}_{ij,aa}^0} \cdot (\mathbf{u}_{j,a} - \mathbf{u}_{i,a}) \quad (93)$$

$$t_a(\mathbf{R}_{ij}^0) \approx t_a(R_{ij}^0) + \nabla_{\mathbf{r}} t_a(\mathbf{r}) \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (94)$$

$$t_{ab}(\mathbf{R}_{ij}^0) \simeq 0 + \nabla_{\mathbf{r}} t_{ab}(\mathbf{r}) \Big|_{\mathbf{R}_{ij}^0} \cdot (\mathbf{u}_j - \mathbf{u}_i) \approx \alpha_{ab} \hat{e}_{ij} \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (95)$$

$$\nabla[t_{ab}(x, y)] = \nabla(xy \cdot f(r)) \quad (96)$$

$$H_e = \sum_i \left(\epsilon_a^{(0)} c_{ai}^\dagger c_{ai} + \epsilon_b^{(0)} c_{bi}^\dagger c_{bi} \right) - \sum_{\langle i, j \rangle} \left(t_a c_{aj}^\dagger c_{ia} + t_b c_{bj}^\dagger c_{ib} + \text{h.c.} \right) + V \sum_i n_{ai} n_{bi} \quad (97)$$

$$H_{ep} = \sum_{\langle i, j \rangle} \left(\alpha_a c_{ja}^\dagger c_{ia} \hat{R}_{ij}^0 + \alpha_b c_{jb}^\dagger c_{ib} \hat{R}_{ij}^0 + \alpha_{ab} c_{ja}^\dagger c_{ib} + \alpha_{ba} c_{jb}^\dagger c_{ia} + \text{h.c.} \right) \cdot (\mathbf{u}_j - \mathbf{u}_i) \quad (98)$$

$$V(\mathbf{R}_i - \mathbf{R}_j) \simeq V_{\text{onsite}} \delta(\mathbf{R}_j - \mathbf{R}_i) + \mathcal{O}(|\mathbf{u}_j - \mathbf{u}_i|) \quad (99)$$

$$\mathbf{u}_j = \frac{1}{\sqrt{N_{\text{uc}}}} \sum_{\mathbf{q}, \lambda} \frac{1}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \xi_{\mathbf{q}\lambda} e^{i\mathbf{q} \cdot \mathbf{R}_j} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \quad (100)$$

$$c_{aj} = \frac{1}{\sqrt{N_{\text{uc}}}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{R}_j} c_{a\mathbf{k}} \quad (101)$$

$$\hat{\delta} = (\delta_x, \delta_y) \in \{\pm \hat{x}, \pm \hat{y}\} \quad (102)$$

$$\sum_{\hat{\delta}} \hat{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) = \hat{x} \cdot (\mathbf{u}_{i+x} - \mathbf{u}_i) - \hat{x} \cdot (\mathbf{u}_{i-x} - \mathbf{u}_i) + \hat{y} \cdot (\mathbf{u}_{i+y} - \mathbf{u}_i) - \hat{y} \cdot (\mathbf{u}_{i-y} - \mathbf{u}_i) \approx 2\nabla \cdot \mathbf{u} \quad (103)$$

$$\sum_{\hat{\delta}} \tilde{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) = (u_{i+x}^y - u_{i-x}^y) - (u_{i+y}^x - u_{i-y}^x) \quad (104)$$

$$H_{e-p} = \sum_{i, \hat{\delta}} \left(D_a c_{ai}^\dagger c_{ai} + D_b c_{bi}^\dagger c_{bi} + \alpha_a c_{a, i+\delta}^\dagger c_{ai} + \alpha_b c_{b, i+\delta}^\dagger c_{bi} \right) \hat{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) + \sum_{i, \hat{\delta}} \left(D_{ab} c_{ai}^\dagger c_{bi} + \text{h.c.} \right) \hat{\delta} \cdot (\mathbf{u}_{i+\delta} - \mathbf{u}_i) \quad (105)$$

$$\mathbf{u}_{i+\delta} - \mathbf{u}_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}, \lambda} \frac{1}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \epsilon_{\mathbf{q}\lambda} \left(e^{i\mathbf{q} \cdot \hat{\delta}} - 1 \right) e^{i\mathbf{q} \cdot \mathbf{R}_i^0} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \quad (106)$$

$$H_{e-p} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}, \lambda} \left[\frac{1}{\sqrt{2M\omega_{\mathbf{q}\lambda}}} \sum_{\hat{\delta}} \hat{\delta} \cdot \epsilon_{\mathbf{q}\lambda} \left(e^{i\mathbf{q} \cdot \hat{\delta}} - 1 \right) e^{i\mathbf{k} \cdot \hat{\delta}} \right] c_{\alpha, \mathbf{k}+\mathbf{q}}^\dagger c_{\beta\mathbf{k}} \left(b_{\mathbf{q}\lambda} + b_{-\mathbf{q}\lambda}^\dagger \right) \quad (107)$$

$$\sum_{\hat{\delta}} \left(e^{i\mathbf{k} \cdot \hat{\delta}} - 1 \right) e^{i\mathbf{k} \cdot \hat{\delta}} \hat{\delta} \cdot \epsilon_{\mathbf{q}\lambda} = \epsilon_{\mathbf{q}\lambda}^x \left[(e^{iq_x} - 1) - (e^{-iq_x} - 1) \right] e^{ik_x} \quad (108)$$

$$g_{aa}^\lambda(\mathbf{k}, \mathbf{q}) = \frac{2i}{\sqrt{2M}} \frac{1}{\sqrt{\omega_{\mathbf{q}\lambda}}} \left[\epsilon_{\mathbf{q}\lambda}^x (D_a \sin(q_x) + \alpha_a (\sin(k_x + q_x) - \sin(k_x))) + \epsilon_{\mathbf{q}\lambda}^y (x \rightarrow y) \right] \quad (109)$$

$$g_{ab}^\lambda(\mathbf{k}, \mathbf{q}) = \frac{2i}{\sqrt{2M}} \frac{1}{\sqrt{\omega_{\mathbf{q}\lambda}}} \left[\epsilon_{\mathbf{q}\lambda}^y (D_{ab} \sin(q_y)) + \epsilon_{\mathbf{q}\lambda}^x (D_{ab} \sin(q_x)) \right] \quad (110)$$

$$\hat{R}_{ij}^0 = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}, \mathbf{q}} \left(c_{a\mathbf{k}}^\dagger, c_{b\mathbf{k}}^\dagger \right) \begin{pmatrix} \alpha_{aa} & \alpha_{ab} \\ \alpha_{ba} & \alpha_{bb} \end{pmatrix} \begin{pmatrix} c_{a, \mathbf{k}+\mathbf{q}} \\ c_{b, \mathbf{k}+\mathbf{q}} \end{pmatrix} \quad (111)$$

$$\rho(\mathbf{r}) = -e \cdot \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}) \quad (112)$$

$$\psi(\mathbf{r}) = \sum_{i,\alpha} \phi_\alpha(\mathbf{r} - \mathbf{R}_i) c_{\alpha i} \quad (113)$$

$$\rho(\mathbf{q}) = -e \sum_{i,j} \sum_{\alpha,\beta} \int d\mathbf{r} e^{-i\mathbf{q}\cdot\mathbf{r}} \phi_\alpha^*(\mathbf{r} - \mathbf{R}_i) \phi_\beta(\mathbf{r} - \mathbf{R}_j) c_{\alpha i}^\dagger c_{\beta j} = -e \sum_{i,\alpha,\beta} e^{-i\mathbf{q}\cdot\mathbf{R}_i} \langle \phi_\alpha | e^{-i\mathbf{q}\cdot\mathbf{r}} | \phi_\beta \rangle c_{\alpha i}^\dagger c_{\beta i} \delta_{ij} \quad (114)$$

$$\rho(\mathbf{q}) \simeq -e \sum_i e^{-i\mathbf{q}\cdot\mathbf{R}_i} \left(n_{ai} + n_{bi} - i\mathbf{q} \cdot 2\langle \phi_a(\mathbf{r}) | \mathbf{r} | \phi_b(\mathbf{r}) \rangle \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} c_{ai} \\ c_{bi} \end{pmatrix} \right) \quad (115)$$

$$\sum_i e^{i\mathbf{q}\cdot\mathbf{R}_i} c_{ai}^\dagger c_{bi} = \frac{1}{N} \sum_{\mathbf{k}} c_{a,\mathbf{k}+\mathbf{q}}^\dagger c_{b\mathbf{k}} \quad (116)$$

2.4 Fotonske sklopitve v resonatorju

Začnemo z minimalnim modelom za elektrone:

$$H_e = H_e^{(0)} + H_V = \sum_k \epsilon_{a,k} c_{a,k}^\dagger c_{a,k} + \sum_k \epsilon_{b,k} c_{b,k}^\dagger c_{b,k} + V \sum_j c_{a,j}^\dagger c_{a,j} c_{b,j}^\dagger c_{b,j} \quad (117)$$

in privzamemo 2D plast, kjer sta dve orbitali na o.c., ki hibridizirata samo preko Coulombske interakcije. Disperzije za model neinteragirajoč sistem so v tem primeru enake (21). Interakcija pa je v prvem redu oblike:

$$H_{e\gamma}^{(1)} = - \int d\mathbf{r} \mathbf{j}_e \cdot \mathbf{A} = - \int d\mathbf{r} \sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) e^{i\mathbf{q}\cdot\mathbf{r}} \mathbf{j}_e(\mathbf{r}) = - \sum_{\mathbf{q}} \mathbf{A}(\mathbf{q}) \cdot \mathbf{j}_e(-\mathbf{q}), \quad (118)$$

kjer je

$$\mathbf{A}(\mathbf{q}) = \sum_{\lambda} \frac{1}{\sqrt{2V\epsilon_0\omega_{\mathbf{q},\lambda}}} \boldsymbol{\epsilon}_{\mathbf{q},\lambda} \left(a_{\mathbf{q},\lambda} + a_{-\mathbf{q},\lambda}^\dagger \right) \quad (119)$$

V t.b. modelu dobimo tok iz kontinuitetne in Heisenbergove enačbe:

$$\frac{\partial \rho(\mathbf{r})}{\partial t} + \nabla \cdot \mathbf{j} = 0 \rightarrow i\mathbf{q} \cdot \mathbf{j}(\mathbf{q}) = -\frac{\partial \rho(\mathbf{q})}{\partial t} \quad (120)$$

(+ je od IBP) in

$$\mathbf{q} \cdot \mathbf{j}_e(\mathbf{q}) = i \frac{\partial \rho(\mathbf{q})}{\partial t} = -[H_e, \rho(\mathbf{q})] \quad (121)$$

Elektronska gostota je

$$\rho(\mathbf{r}) = -e \cdot \psi^\dagger(\mathbf{r}) \psi(\mathbf{r}) \quad (122)$$

operatorje polja razvijemo po lokaliziranih / Wannierovih orbitalah

$$\psi(\mathbf{r}) = \sum_{i,\gamma=\{a,b\}} \phi_\gamma(\mathbf{r} - \mathbf{R}_i) c_{\gamma i}. \quad (123)$$

Transformiramo s FT, uvedemo $\mathbf{r} = \mathbf{R}_i + \mathbf{x}$ in razvijemo do dipolov $e^{-i\mathbf{q}\cdot\mathbf{x}} \simeq 1 - i\mathbf{q} \cdot \mathbf{x}$ (delamo se da se orbitale ne prekrivajo izven o.c.):

$$\rho(\mathbf{q}) = -e \sum_{i,j} \sum_{\gamma,\tau} \int d\mathbf{r} e^{-i\mathbf{q}\cdot\mathbf{r}} \phi_\gamma^*(\mathbf{r} - \mathbf{R}_i) \phi_\tau(\mathbf{r} - \mathbf{R}_j) c_{\gamma i}^\dagger c_{\tau j} = -e \sum_{i,\gamma,\tau} e^{-i\mathbf{q}\cdot\mathbf{R}_i} \langle \phi_\gamma | e^{-i\mathbf{q}\cdot\mathbf{x}} | \phi_\tau \rangle c_{\gamma i}^\dagger c_{\tau i} \quad (124)$$

Zdaj pa recimo da je dipolni moment samo med različnima orbitalama v o.c. in $n_{\gamma i} = \langle \phi_\gamma | \phi_\gamma \rangle c_{\gamma i}^\dagger c_{\gamma i}$:

$$\rho(\mathbf{q}) \simeq -e \sum_i e^{-i\mathbf{q}\cdot\mathbf{R}_i} \left(n_{ai} + n_{bi} - i\mathbf{q} \cdot \mathbf{d} \left(c_{a,i}^\dagger c_{b,i} + c_{b,i}^\dagger c_{a,i} \right) \right) \quad (125)$$

zdaj uporabimo zvezo

$$\sum_i e^{i\mathbf{q}\cdot\mathbf{R}_i} c_{ai}^\dagger c_{bi} = \sum_{\mathbf{k}} c_{a,\mathbf{k}+\mathbf{q}}^\dagger c_{b\mathbf{k}} \quad (126)$$

z negativnim argumentom je potem elektronska gostota do dipolnega člena (upoštevajoč $\mathbf{d} = \langle \phi_a(\mathbf{x}) | \mathbf{x} | \phi_b(\mathbf{x}) \rangle$):

$$\rho(-\mathbf{q}) = -e \sum_{\mathbf{k}} \begin{pmatrix} c_{a,\mathbf{k}+\mathbf{q}}^\dagger & c_{b,\mathbf{k}+\mathbf{q}}^\dagger \end{pmatrix} \left(\mathbb{1} + i \mathbf{q} \cdot \mathbf{d} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \right) \begin{pmatrix} c_{a\mathbf{k}} \\ c_{b\mathbf{k}} \end{pmatrix} \quad (127)$$

Zdaj poračunamo komutator, da dobimo tok. **Lahko vzamemo H_e^0 , ker za local V se členi z $\Delta_k \rightarrow \Delta$ pokrajsajo!!**, načeloma pa (nisem še preveru, z dipolnim členom ne komutira). Delam z vektorji $\psi_{\mathbf{k}} = \begin{pmatrix} c_{a,\mathbf{k}} & c_{b,\mathbf{k}} \end{pmatrix}^\top$:

$$[H_e^{(0)}, \rho(\mathbf{q})] = -e \sum_{\mathbf{k}, \mathbf{k}'} \left[\psi_{\mathbf{k}}^\dagger \begin{pmatrix} \varepsilon_{a,\mathbf{k}} & 0 \\ 0 & \varepsilon_{b,\mathbf{k}} \end{pmatrix} \psi_{\mathbf{k}}, \psi_{\mathbf{k}'-\mathbf{q}}^\dagger \begin{pmatrix} 1 & -i\mathbf{q} \cdot \mathbf{d} \\ -i\mathbf{q} \cdot \mathbf{d} & 1 \end{pmatrix} \psi_{\mathbf{k}'} \right] \quad (128)$$

uporabimo zvezo:

$$\left[\psi_{\mathbf{k}}^\dagger A \psi_{\mathbf{k}}, \psi_{\mathbf{k}'-\mathbf{q}}^\dagger B \psi_{\mathbf{k}'} \right] = \psi_{\mathbf{k}'-\mathbf{q}}^\dagger \left(A(\mathbf{k}' - \mathbf{q}) B(\mathbf{k}') - B(\mathbf{k}' - \mathbf{q}) A(\mathbf{k}') \right) \psi_{\mathbf{k}'}, \delta_{\mathbf{k}, \mathbf{k}'-\mathbf{q}} \quad (129)$$

$$= \psi_{\mathbf{k}}^\dagger \left(A(\mathbf{k}) B(\mathbf{k} + \mathbf{q}) - B(\mathbf{k}) A(\mathbf{k} + \mathbf{q}) \right) \psi_{\mathbf{k}+\mathbf{q}} \quad (130)$$

od tod

$$\mathbf{q} \cdot \mathbf{j}_e(\mathbf{q}) = -[H_e^{(0)}, \rho(\mathbf{q})] = -e \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger \begin{pmatrix} \varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}} & i(\mathbf{q} \cdot \mathbf{d})(\varepsilon_{a,\mathbf{k}} - \varepsilon_{b,\mathbf{k}+\mathbf{q}}) \\ i(\mathbf{q} \cdot \mathbf{d})(\varepsilon_{b,\mathbf{k}} - \varepsilon_{a,\mathbf{k}+\mathbf{q}}) & \varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}} \end{pmatrix} \psi_{\mathbf{k}+\mathbf{q}} \quad (131)$$

da izrazimo tok, naredimo približek, da prispevajo le majhni $\mathbf{q} \rightarrow 0$, zato:

$$\varepsilon_{a\mathbf{k}} - \varepsilon_{a,\mathbf{k}+\mathbf{q}} \simeq -\mathbf{q} \cdot \nabla_{\mathbf{k}} \varepsilon_{a,\mathbf{k}+\mathbf{q}/2}, \quad (132)$$

kar drži točno za proste elektrone in približno za tight binding disperzije. In general ce ne expandamo:

$$\mathbf{j}_e(\mathbf{q}) = -\frac{e\mathbf{q}}{\|\mathbf{q}\|^2} \sum_{\mathbf{k}} \psi_{\mathbf{k}}^\dagger M(\mathbf{k}, \mathbf{q}) \psi_{\mathbf{k}+\mathbf{q}} + \mathbf{q} \times \mathbf{C}(\mathbf{q}) \quad (133)$$

kjer so $\mathbf{C}(\mathbf{q})$ arbitrarna vektorska polja (tokovi brez divergence, magnetizacijski tokovi). Ce pa naredimo ta približek pa dobimo za tok v splošnem, indekse zamaknemo za $-\mathbf{q}$:

$$\mathbf{j}_e(\mathbf{q}) \simeq -e \sum_{\mathbf{k}} \psi_{\mathbf{k}-\mathbf{q}}^\dagger \begin{pmatrix} \nabla_{\mathbf{k}} \varepsilon_{a,\mathbf{k}-\mathbf{q}/2} & i\mathbf{d}(\varepsilon_{a,\mathbf{k}-\mathbf{q}} - \varepsilon_{b\mathbf{k}}) \\ i\mathbf{d}(\varepsilon_{b,\mathbf{k}-\mathbf{q}} - \varepsilon_{a\mathbf{k}}) & \nabla_{\mathbf{k}} \varepsilon_{b,\mathbf{k}-\mathbf{q}/2} \end{pmatrix} \psi_{\mathbf{k}}. \quad (134)$$

Od tod splošna Hamiltonka v prvem redu:

$$H_{e\gamma}^{(1)} = - \sum_{\mathbf{q}} \mathbf{j}_e(-\mathbf{q}) \cdot \mathbf{A}(\mathbf{q}) = \quad (135)$$

$$= \sum_{\mathbf{k}, \mathbf{q}, \lambda} \frac{e}{\sqrt{2\mathcal{V}\varepsilon_0\omega_{\mathbf{q},\lambda}}} \psi_{\mathbf{k}+\mathbf{q}}^\dagger \underbrace{\begin{pmatrix} \varepsilon_{\mathbf{q}\lambda} \cdot \mathbf{v}_a(\mathbf{k} + \frac{\mathbf{q}}{2}) & i(\varepsilon_{\mathbf{q}\lambda} \cdot \mathbf{d})(\varepsilon_{a,\mathbf{k}+\mathbf{q}} - \varepsilon_{b,\mathbf{k}}) \\ i(\varepsilon_{\mathbf{q}\lambda} \cdot \mathbf{d})(\varepsilon_{b,\mathbf{k}+\mathbf{q}} - \varepsilon_{a,\mathbf{k}}) & \varepsilon_{\mathbf{q}\lambda} \cdot \mathbf{v}_b(\mathbf{k} + \frac{\mathbf{q}}{2}) \end{pmatrix}}_{M^\lambda(\mathbf{k}, \mathbf{q})} \psi_{\mathbf{k}} \left(a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^\dagger \right). \quad (136)$$

Za setup od Curtisa: material 2D v xy ravnini, stoječe EM valovanje v z , vzamemo osnovni mode ($n = 1$), ki prispeva največ, v tem primeru sta dve možni polarizaciji:

$$\epsilon_{\text{TE}}(z) = (\hat{\mathbf{z}} \times \hat{\mathbf{q}}) \sin\left(\frac{\pi}{L}z\right) \quad (137)$$

$$\epsilon_{\text{TM}}(z) = \frac{\omega_0}{\omega_{\mathbf{q}}} \hat{\mathbf{q}} \sin\left(\frac{\pi}{L}z\right) + i \frac{c}{\omega_{\mathbf{q}}} \hat{\mathbf{z}} \cos\left(\frac{\pi}{L}z\right) \quad (138)$$

kjer je $\mathbf{q} = q(\cos \varphi, \sin \varphi)$ valovni vektor v ravnini, $\omega_0 = \pi c/L$ in $\omega_{\mathbf{q}} = \sqrt{\omega_0^2 + c^2 q^2}$. Na sredini resonatorja, pri $z = L/2$, je $\sin(\pi z/L) = 1$ in $\cos(\pi z/L) = 0$, zato:

$$\epsilon_{\text{TE}}\left(\frac{L}{2}\right) = \hat{\mathbf{z}} \times \hat{\mathbf{q}} = \begin{pmatrix} -\sin \varphi \\ \cos \varphi \\ 0 \end{pmatrix} \quad (139)$$

$$\epsilon_{\text{TM}}\left(\frac{L}{2}\right) = \frac{\omega_0}{\omega_{\mathbf{q}}} \hat{\mathbf{q}} = \frac{\omega_0}{\omega_{\mathbf{q}}} \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{pmatrix} \quad (140)$$

Iz tega imamo pri $z = L/2$ možne sklopitve glede na model ($\mathbf{d} = d(\cos \theta, \sin \theta)$ v ravnini oz. $\mathbf{d} = d\hat{\mathbf{z}}$ izven ravnine):

Mode	$\epsilon \cdot \mathbf{v}_{\gamma}\left(\mathbf{k} + \frac{\mathbf{q}}{2}\right)$	$\epsilon \cdot \mathbf{d}, \mathbf{d}$ in-plane	$\epsilon \cdot \mathbf{d}, \mathbf{d} \parallel \hat{\mathbf{z}}$
TE	$2t_{\gamma} \left(-\sin \varphi \sin \left(k_x + \frac{q_x}{2} \right) + \cos \varphi \sin \left(k_y + \frac{q_y}{2} \right) \right)$	$d \sin(\theta - \varphi)$	0
TM	$\frac{\omega_0}{\omega_{\mathbf{q}}} 2t_{\gamma} \left(\cos \varphi \sin \left(k_x + \frac{q_x}{2} \right) + \sin \varphi \sin \left(k_y + \frac{q_y}{2} \right) \right)$	$\frac{\omega_0}{\omega_{\mathbf{q}}} d \cos(\theta - \varphi)$	$i d \frac{c q}{\omega_{\mathbf{q}}} \cos\left(\frac{\pi}{L} z\right) (= 0 \text{ pri } z = \frac{L}{2})$

dipolni moment je neničeln za liho-sode kombinacije orbital in plane, tega gledamo samo glede na o.c.:

$$\mathbf{d} = \langle \phi_a | \mathbf{x} | \phi_b \rangle = \int \phi_a^*(\mathbf{x}) \mathbf{x} \phi_b(\mathbf{x}) d\mathbf{x} \quad (141)$$

Kaj pa če gledamo namesto tight binding, proste elektrone, torej

$$\epsilon_{\gamma\mathbf{k}} = \pm \frac{G}{2} \mp 2t(\cos k_x + \cos k_y) \quad \text{za } k_x, k_y \sim 0 \quad (142)$$

od tod

$$\epsilon_{\gamma\mathbf{k}} \approx \pm \frac{G}{2} \mp 2t \left(1 - \frac{k_x^2}{2} + 1 - \frac{k_y^2}{2} \right) = \pm \left(\frac{G}{2} - 4t \right) \pm t(k_x^2 + k_y^2) = \pm(\epsilon^0 + t\mathbf{k} \cdot \mathbf{k}) \quad (143)$$

Torej je Hamiltonka

$$H_e = \sum_{\gamma, \mathbf{k}} (\epsilon_{\gamma}^0 + t_{\gamma} \mathbf{k}^2) c_{\gamma, \mathbf{k}}^{\dagger} c_{\gamma, \mathbf{k}} \quad (144)$$

Na drug način lahko poračunamo tok, ker razvijemo

$$\psi(\mathbf{r}) = \frac{1}{\sqrt{\mathcal{V}}} \sum_{\mathbf{k}, \gamma=\{a, b\}} e^{i\mathbf{k} \cdot \mathbf{r}} c_{\gamma, \mathbf{k}}. \quad (145)$$

in za tok uporabimo

$$\mathbf{j}_e(\mathbf{r}) = -\frac{e}{2mi} (\psi^{\dagger}(\mathbf{r}) \nabla \psi(\mathbf{r}) - \nabla \psi^{\dagger}(\mathbf{r}) \psi(\mathbf{r})) = -\frac{e}{2mi} \frac{1}{\mathcal{V}} \sum_{\gamma, \mathbf{k}, \mathbf{k}'} (c_{\gamma \mathbf{k}'}^{\dagger} c_{\gamma \mathbf{k}} e^{-i\mathbf{k}' \cdot \mathbf{r}} (i\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} - c_{\gamma \mathbf{k}'}^{\dagger} c_{\gamma \mathbf{k}} (-i\mathbf{k}') e^{-i\mathbf{k}' \cdot \mathbf{r}} e^{i\mathbf{k} \cdot \mathbf{r}}) \quad (146)$$

s FT dobimo

$$\mathbf{j}_e(-\mathbf{q}) = \int d\mathbf{r} \mathbf{j}_e(\mathbf{r}) e^{i\mathbf{q} \cdot \mathbf{r}} = -\frac{e}{2m} \sum_{\gamma, \mathbf{k}} (c_{\gamma, \mathbf{k}+\mathbf{q}}^{\dagger} c_{\gamma \mathbf{k}} \mathbf{k} + (\mathbf{k} + \mathbf{q}) c_{\gamma, \mathbf{k}+\mathbf{q}}^{\dagger} c_{\gamma \mathbf{k}}) = -\frac{e}{m} \sum_{\mathbf{k}} \left(\mathbf{k} + \frac{\mathbf{q}}{2} \right) \psi_{\mathbf{k}+\mathbf{q}}^{\dagger} \psi_{\mathbf{k}} \quad (147)$$

kar bi dobili tudi iz bolj splošne relacije. V tem primeru dobimo enako kot Curtis, $m = 1/2t$

$$H_{e\gamma}^{(1)} = \frac{e}{m} \sum_{\mathbf{k}, \mathbf{q}, \lambda} \frac{1}{\sqrt{2\mathcal{V}\epsilon_0\omega_{\mathbf{q}, \lambda}}} \left(\mathbf{k} + \frac{\mathbf{q}}{2} \right) \cdot \epsilon_{\mathbf{q}\lambda} \psi_{\mathbf{k}+\mathbf{q}}^{\dagger} \psi_{\mathbf{k}} (a_{\mathbf{q}\lambda} + a_{-\mathbf{q}\lambda}^{\dagger}) \quad (148)$$

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